

---

# Patient Management System

## User Manual

---

A comprehensive guide to using the PMS web platform —  
patient records, appointments, prescriptions, lab results, and billing

**Version 1.0**

**Last Updated:** September 2026

**Department of Computer Engineering**  
University of Peradeniya

September 25, 2026

# Contents

|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                 | <b>3</b>  |
|          | Key Benefits . . . . .                              | 3         |
| <b>2</b> | <b>Getting Started</b>                              | <b>4</b>  |
| 2.1      | System Requirements . . . . .                       | 4         |
| 2.2      | Accessing the Application . . . . .                 | 4         |
| <b>3</b> | <b>Signing Up &amp; Logging In</b>                  | <b>5</b>  |
| 3.1      | Creating a New Account . . . . .                    | 5         |
| 3.2      | Logging In . . . . .                                | 6         |
| 3.3      | Session Management . . . . .                        | 6         |
| 3.4      | Logging Out . . . . .                               | 7         |
| <b>4</b> | <b>Understanding User Roles</b>                     | <b>8</b>  |
| <b>5</b> | <b>Patient Dashboard</b>                            | <b>9</b>  |
| 5.1      | Overview . . . . .                                  | 9         |
| 5.2      | My Appointments . . . . .                           | 9         |
| 5.3      | Medical Records . . . . .                           | 9         |
| 5.4      | Profile Management . . . . .                        | 10        |
| 5.5      | Billing & Invoices . . . . .                        | 10        |
| <b>6</b> | <b>Doctor Dashboard</b>                             | <b>11</b> |
| 6.1      | Overview . . . . .                                  | 11        |
| 6.2      | Appointments . . . . .                              | 11        |
| 6.3      | Patient Records . . . . .                           | 11        |
| 6.4      | Prescriptions . . . . .                             | 12        |
| <b>7</b> | <b>Nurse Dashboard</b>                              | <b>13</b> |
| 7.1      | Patient Vitals . . . . .                            | 13        |
| 7.2      | Appointment Assistance . . . . .                    | 13        |
| 7.3      | Patient Information . . . . .                       | 13        |
| <b>8</b> | <b>Receptionist Dashboard</b>                       | <b>14</b> |
| 8.1      | Patient Registration . . . . .                      | 14        |
| 8.2      | Appointment Management . . . . .                    | 14        |
| 8.3      | Patient Queue . . . . .                             | 14        |
| 8.4      | Front-Desk Billing & Consultation Charges . . . . . | 15        |
| <b>9</b> | <b>Pharmacist Dashboard</b>                         | <b>16</b> |
| 9.1      | Prescriptions Queue . . . . .                       | 16        |
| 9.2      | Medication Management . . . . .                     | 16        |

|                                                          |           |
|----------------------------------------------------------|-----------|
| <b>10 Lab Technician Dashboard</b>                       | <b>17</b> |
| 10.1 Lab Orders . . . . .                                | 17        |
| 10.2 Results Entry & Automated Charge Dispatch . . . . . | 17        |
| 10.3 Report Management . . . . .                         | 17        |
| <b>11 Billing Staff Dashboard</b>                        | <b>18</b> |
| 11.1 Centralized Billing Workflow Overview . . . . .     | 18        |
| 11.2 Process Pending Bills Panel . . . . .               | 19        |
| 11.3 Invoice Management & Invoices List . . . . .        | 19        |
| 11.4 Payment Processing . . . . .                        | 20        |
| 11.5 Revenue Summary & Financial Metrics . . . . .       | 20        |
| <b>12 Admin Dashboard</b>                                | <b>21</b> |
| 12.1 User Management . . . . .                           | 21        |
| 12.2 Audit Logs . . . . .                                | 21        |
| 12.3 System Reports . . . . .                            | 22        |
| <b>13 Management Dashboard</b>                           | <b>23</b> |
| 13.1 Staff Management . . . . .                          | 23        |
| 13.2 Operational Reports . . . . .                       | 23        |
| <b>14 Notifications</b>                                  | <b>24</b> |
| <b>15 Frequently Asked Questions (FAQ)</b>               | <b>25</b> |
| <b>16 Troubleshooting</b>                                | <b>26</b> |
| <b>17 Contact &amp; Support</b>                          | <b>27</b> |
| Project Links . . . . .                                  | 27        |

## Chapter 1

# Introduction

The **Patient Management System (PMS)** is a web-based application designed to streamline healthcare operations by centralizing patient records, appointment scheduling, medical history, prescriptions, lab results, and billing into a single, easy-to-use platform.

## Key Benefits

- **Centralized Records** — All patient data in one place, accessible to authorized staff.
- **Role-Based Access** — Each user sees only the features and data relevant to their role.
- **Real-Time Notifications** — Instant updates via WebSocket-powered notifications.
- **Google Sign-In** — Quick and secure login with your Google account.
- **Secure & Compliant** — JWT authentication, encrypted passwords, and audit logging.

## Chapter 2

# Getting Started

## 2.1 System Requirements

| Requirement     | Details                                                                       |
|-----------------|-------------------------------------------------------------------------------|
| <b>Browser</b>  | Google Chrome (v100+), Firefox (v100+), Microsoft Edge (v100+), Safari (v15+) |
| <b>Internet</b> | Stable internet connection                                                    |
| <b>Screen</b>   | Minimum 1024 × 768 resolution (responsive design supports mobile)             |

## 2.2 Accessing the Application

1. Open your web browser.
2. Navigate to the application URL provided by your administrator (e.g., <https://e22-co2060-patient-management-syste-three.vercel.app>).
3. You will be presented with the **Home Page** featuring an overview of the system.

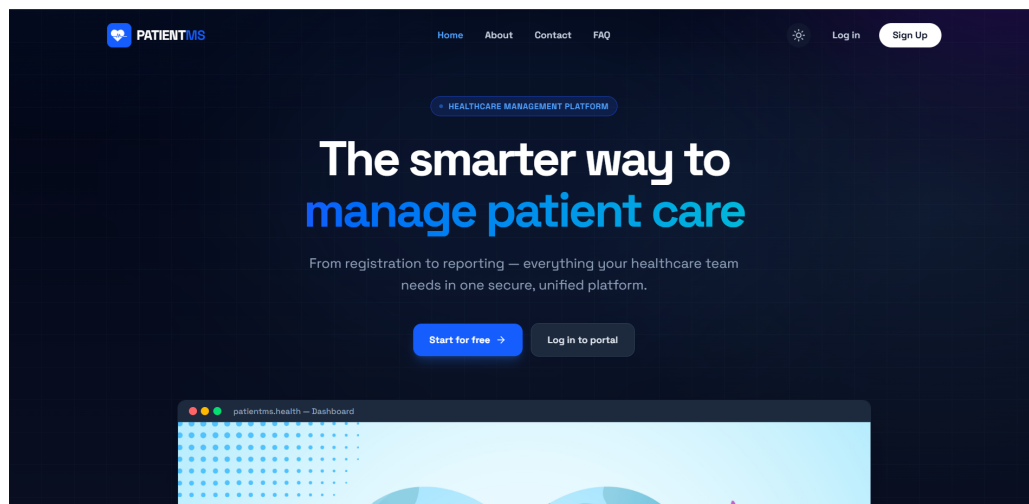


Figure 2.1: The PMS Home page

## Chapter 3

# Signing Up & Logging In

### 3.1 Creating a New Account

1. Click “**Sign Up**” or “**Start with free**” from the navigation bar or home page.

**Smarter healthcare management.**

Create your account and start managing hospital operations smarter from day one.

- ✓ Real-time patient triage
- ✓ Integrated lab management
- ✓ Automated billing & invoicing
- ✓ Role-based access control

Already have an account? [Sign in here](#)

**Create your account**

Already registered? [Sign in](#)

[Sign up with Google](#)

or register with email

**First name**  **Last name**

**Email address**

**Mobile number**

**Password**

**Confirm password**

[Create account →](#)

By signing up you agree to our [Terms of Service](#) and [Privacy Policy](#).

Figure 3.1: The PMS SignUp page

2. Fill in the required fields:
  - **First Name** and **Last Name**
  - **Email Address** (must be unique)
  - **Mobile Number**
  - **Password** (must meet security requirements)
  - **Role** — The default role is Patient. Other administrative and clinical roles are appointed by system administrators (e.g., Doctor, Nurse, Receptionist, Pharmacist, Lab Technician, Billing Staff).
3. Alternatively, click “**Sign in with Google**” to use your Google account.
4. Click “**Sign Up**” to create your account.

#### Note

Some roles (e.g., Doctor, Nurse, Admin) may require administrator approval before access is granted.

## 3.2 Logging In

1. Click “**Login**” from the navigation bar.
2. Enter your **Email** and **Password**.
3. Alternatively, use “**Sign in with Google**” for one-click login.

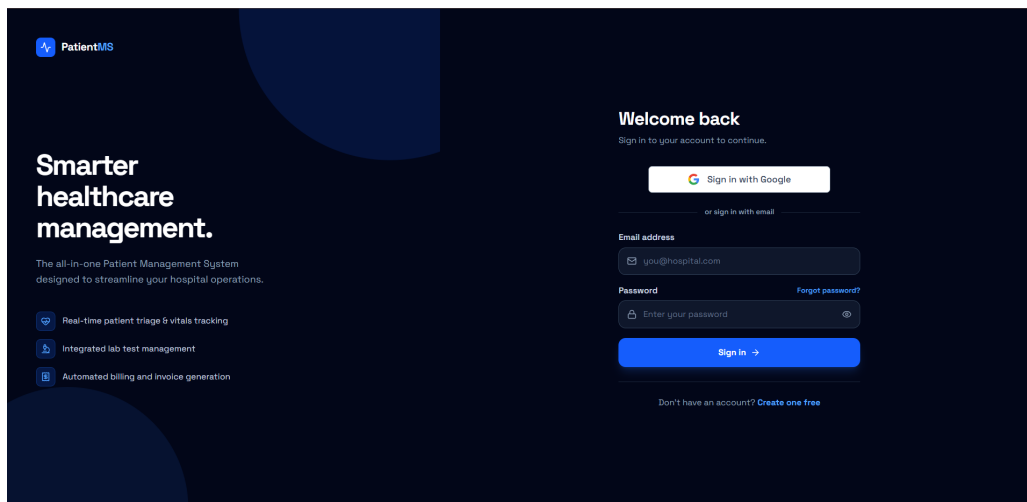


Figure 3.2: The PMS login page

4. Upon successful login, you will be automatically redirected to your role-specific dashboard.

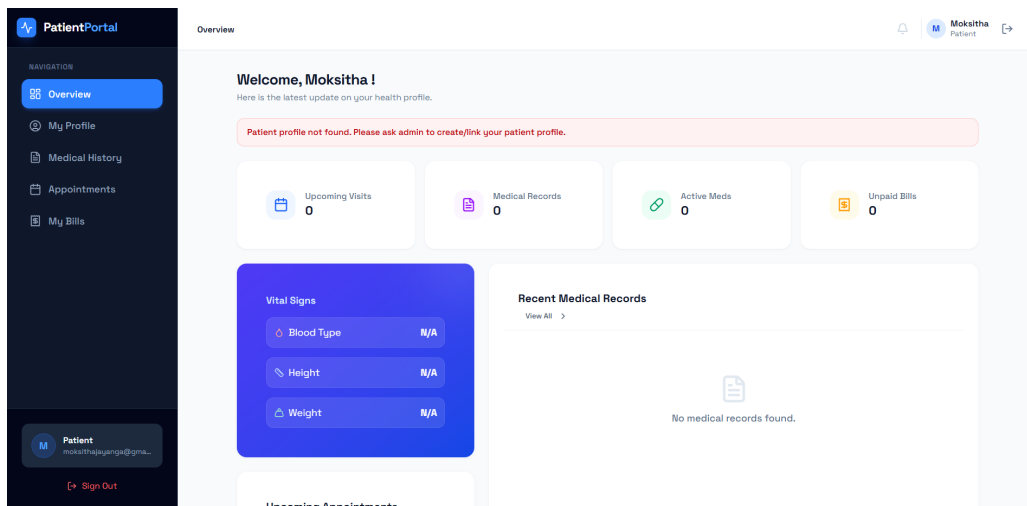


Figure 3.3: The PMS Profile page

## 3.3 Session Management

- Your session is secured with **JWT tokens** that expire after **15 minutes**.
- The system automatically refreshes your session using a **refresh token** (valid for 7 days).
- If your session expires completely, you will be redirected to the login page.

## 3.4 Logging Out

- Click the **Logout** button in the navigation bar or sidebar.
- This invalidates your session on both the browser and the server.

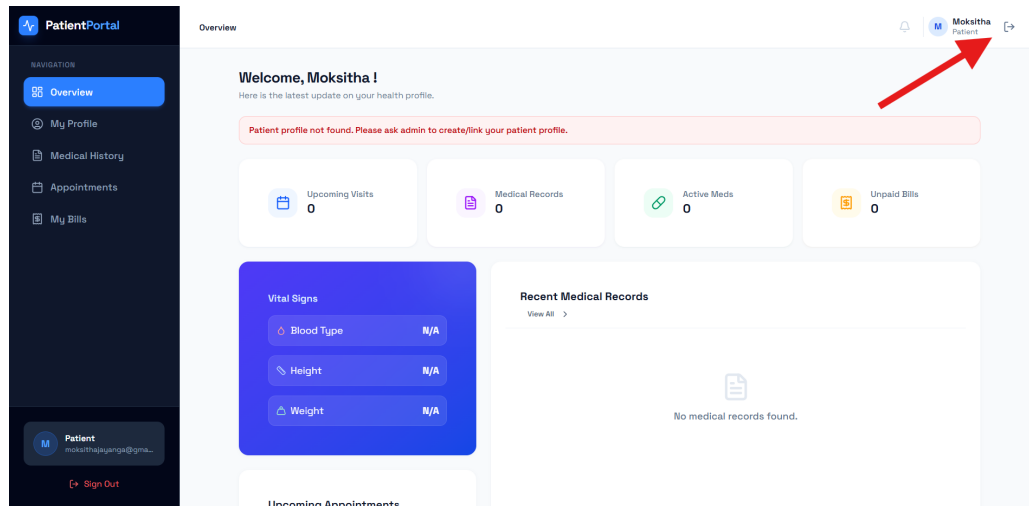


Figure 3.4: The PMS logout page

## Chapter 4

# Understanding User Roles

The PMS uses **role-based access control (RBAC)**. Each role has access to specific features:

| Role                  | Key Permissions                                                                                                               |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Patient</b>        | View own medical records, appointments, prescriptions, lab results, and itemized billing/invoices                             |
| <b>Doctor</b>         | Full clinical access — diagnoses, prescriptions, medical records, appointments                                                |
| <b>Nurse</b>          | Record vitals, assist doctors, view patient information                                                                       |
| <b>Receptionist</b>   | Book/manage appointments, register patients, queue consultation fees, front-desk invoicing & payments                         |
| <b>Pharmacist</b>     | View and manage prescriptions, dispense medications, auto-queue medication charges to billing                                 |
| <b>Lab Technician</b> | Enter and manage lab test results, upload reports, auto-queue lab diagnostic fees to billing                                  |
| <b>Billing Staff</b>  | Centralized billing hub — process pending department charges, issue consolidated invoices, record payments, revenue analytics |
| <b>Admin</b>          | Manage users, view audit logs, system reports                                                                                 |
| <b>Super Admin</b>    | Full system access, manage all users and configurations                                                                       |
| <b>Management</b>     | User & staff management, administrative reporting                                                                             |

## Chapter 5

# Patient Dashboard

After logging in as a **Patient** (after account approval by management), you will see your personalized dashboard:

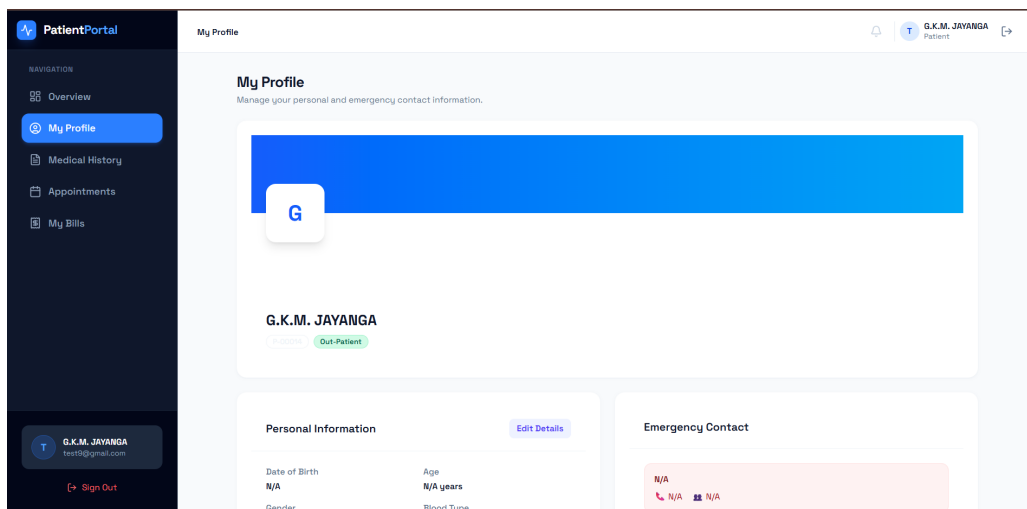


Figure 5.1: The PMS Patient Profile page

## 5.1 Overview

Summary cards showing upcoming appointments, recent prescriptions, and pending lab results.

## 5.2 My Appointments

- **View** all your scheduled, completed, and cancelled appointments.
- **Book New Appointment** — Select a doctor, choose a date/time, and provide the reason for visit.
- **Cancel Appointment** — Cancel an upcoming appointment if needed.

## 5.3 Medical Records

View your complete medical history including:

- **Diagnoses** — Past and current conditions
- **Prescriptions** — Medications prescribed by your doctors

- **Lab Results** — Test results and reports
- **Clinical Notes** — Notes from your healthcare providers

## 5.4 Profile Management

View and update your personal information:

- Contact details
- Emergency contact information
- Medical history and allergies

## 5.5 Billing & Invoices

- **View Invoices** — Access past and current invoices issued by the healthcare facility.
- **Itemized Charge Breakdown** — Review specific line items categorized by department:
  - Pharmacy charges (prescribed medications, dosage quantities, unit prices)
  - Laboratory charges (tests conducted, diagnostic panel fees)
  - Consultation and clinical procedure fees
- **Payment Status Tracking** — Check invoice payment status (ISSUED, PARTIALLY\_PAID, PAID).
- **View Payment History & Receipts** — Inspect payment amounts, dates, payment methods, and remaining balance.

## Chapter 6

# Doctor Dashboard

The **Doctor Dashboard** provides full clinical tools:

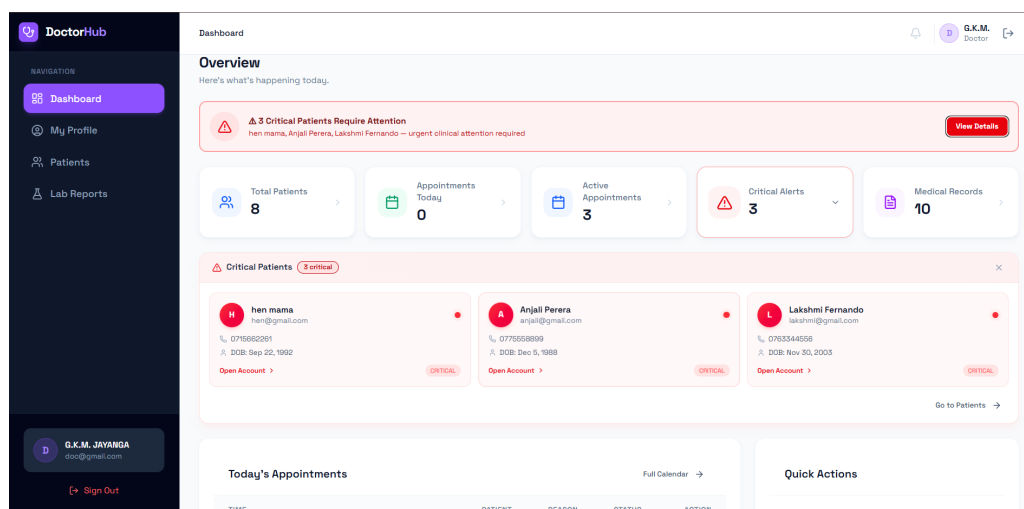


Figure 6.1: The PMS Doctor Dashboard

## 6.1 Overview

Summary of today's appointments, pending reviews, and patient queue.

## 6.2 Appointments

- View daily/weekly schedule of patient appointments.
- Mark appointments as **Completed** or **No-Show**.
- Add clinical notes during or after appointments.

## 6.3 Patient Records

- Search and view patient profiles and medical history.
- **Create Medical Records** — Add new diagnoses, prescriptions, lab orders, and clinical notes.
- **Update Records** — Modify existing medical records.

Record types supported:

- **DIAGNOSIS** — Medical conditions and findings
- **PRESCRIPTION** — Medications and dosage instructions
- **LAB\_RESULT** — Laboratory test orders and results
- **IMAGING** — Radiology and imaging reports

## 6.4 Prescriptions

- Write new prescriptions with medication details, dosage, and duration.
- View prescription history for any patient.

## Chapter 7

# Nurse Dashboard

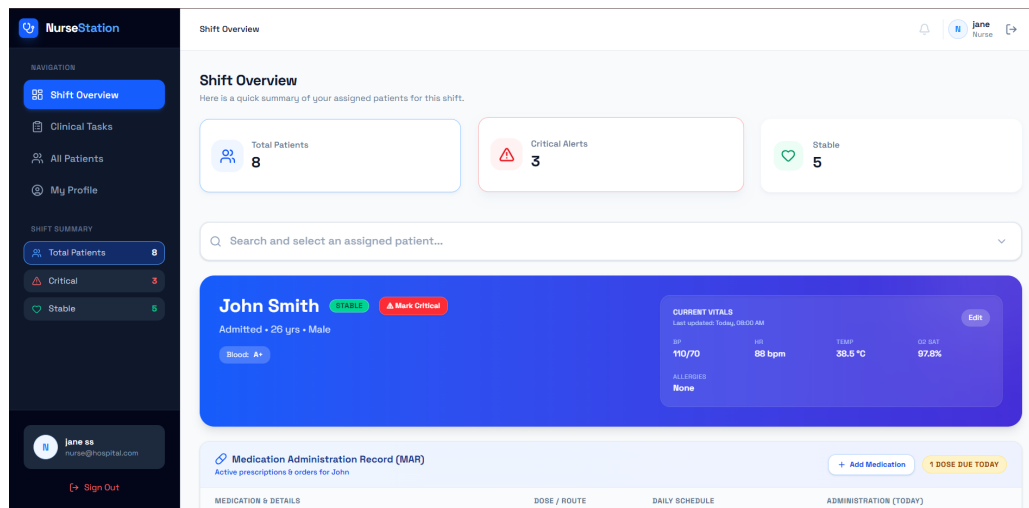


Figure 7.1: The PMS Nurse Dashboard

The **Nurse Dashboard** assists nurses in clinical support tasks:

### 7.1 Patient Vitals

- Record patient vital signs (blood pressure, temperature, pulse, etc.).
- View patient queues for the day.

### 7.2 Appointment Assistance

- View upcoming appointments assigned to doctors.
- Help prepare patient information before consultations.

### 7.3 Patient Information

Access patient profiles and relevant medical history for care coordination.

## Chapter 8

# Receptionist Dashboard

The **Receptionist Dashboard** focuses on front-desk operations:

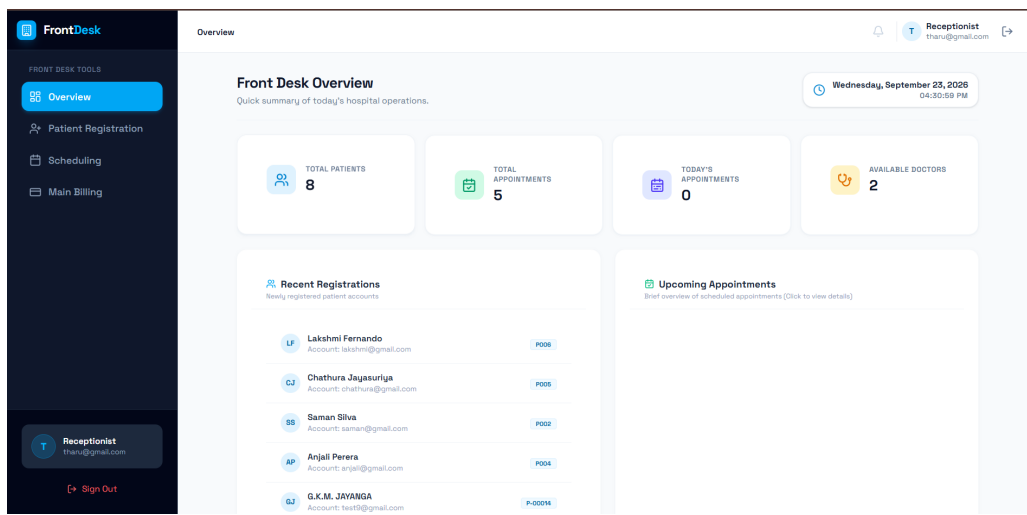


Figure 8.1: The PMS Receptionist Dashboard

## 8.1 Patient Registration

- Register new patients entering the facility.
- Search for existing patient records.

## 8.2 Appointment Management

- **Schedule** new appointments for patients with available doctors.
- **Reschedule** or **Cancel** existing appointments.
- View doctor availability and schedules.

## 8.3 Patient Queue

Manage the daily patient check-in and waiting queue.

## 8.4 Front-Desk Billing & Consultation Charges

- **Queue Consultation Charges** — Dispatch consultation fees directly to the patient's pending bill queue.
- **Direct Invoice Generation** — Issue consolidated invoices for walk-in visits or appointment check-outs.
- **Process Payments** — Accept payments at the front desk via Cash, Card, or Insurance.

## Chapter 9

# Pharmacist Dashboard

The **Pharmacist Dashboard** manages prescription workflows:

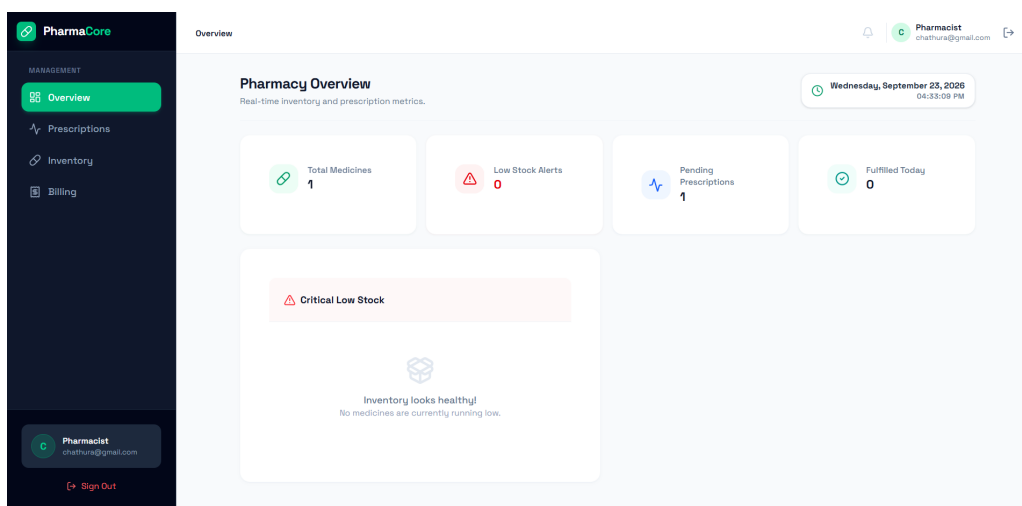


Figure 9.1: The PMS Pharmacist Dashboard

## 9.1 Prescriptions Queue

- View incoming prescriptions from doctors.
- Mark prescriptions as **Dispensed** or **Pending**.
- **Automated Billing Staging** — When medications are dispensed, itemized medication costs (unit price  $\times$  quantity) are automatically dispatched to the billing system's pending bill queue (`pending_bill_items`), eliminating manual reentry.

## 9.2 Medication Management

- Search medication inventory and pricing.
- Track dispensing history.

## Chapter 10

# Lab Technician Dashboard

The **Lab Technician Dashboard** handles laboratory operations:

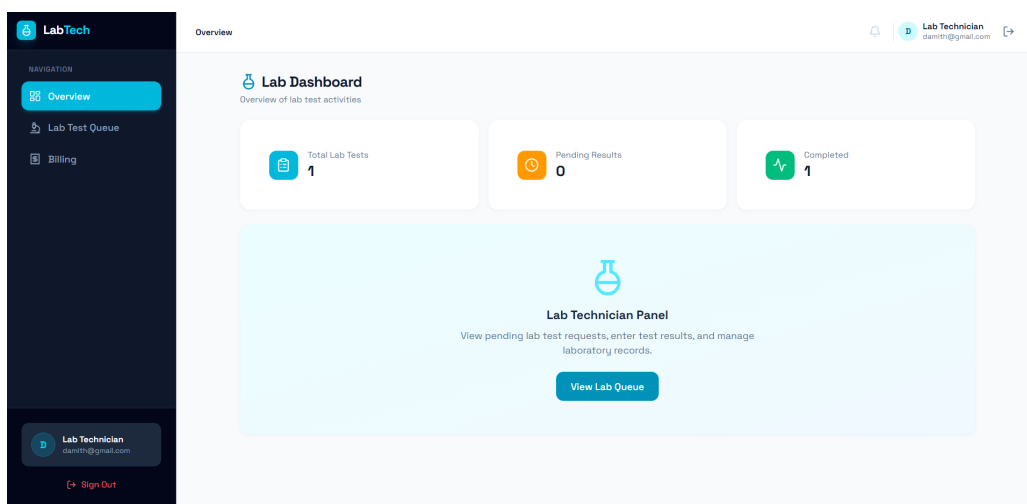


Figure 10.1: The PMS Lab Technician Dashboard

## 10.1 Lab Orders

- View pending lab test orders from doctors.
- Update test status (Pending → In Progress → Completed).

## 10.2 Results Entry & Automated Charge Dispatch

- Enter lab test results and attach reports.
- Upload lab report files (PDF, images up to 10 MB).
- **Automated Billing Staging** — Completing a lab order automatically forwards the corresponding diagnostic test charges to the patient's pending billing queue.

## 10.3 Report Management

- View and manage completed lab reports.
- Reports are accessible to the ordering doctor and the patient.

## Chapter 11

# Billing Staff Dashboard

The **Billing Staff Dashboard** serves as the central financial hub of the hospital, integrating service charges across all clinical departments into consolidated invoices and managing the revenue lifecycle.

The screenshot displays the 'Generate Invoice' interface within the BillingHub system. The interface is divided into three main sections for creating a bill:

- 1. Select Patient:** Includes a search bar labeled 'Search patients by name, NIC'.
- 2. Invoice Items:** A table for adding items to the invoice.

| ITEM DESCRIPTION      | TYPE     | QTY | UNIT PRICE | AMOUNT |
|-----------------------|----------|-----|------------|--------|
| e.g. Consultation Fee | Consulta | 1   | 0.00       | 0.00   |

Below the table is a '+ Add Item' button. To the right, a summary box shows: Subtotal: Rs. 0.00, Discount: - Rs. 0.00, Tax: + Rs. 0.00, and Total: Rs. 0.00.
- 3. Adjustments & Notes:** Includes input fields for 'Discount (Rs.)' and 'Tax (Rs.)', both currently set to 0. It also has a section for 'Internal Notes' with a text area labeled 'Additional billing notes...'.

At the bottom right, there are 'Cancel' and 'Generate Invoice' buttons. The left sidebar shows the 'BillingHub' logo, 'REVENUE & BILLING' menu, 'Billing & Invoices' sub-menu, and a 'Create Invoice' button. The user profile at the bottom left is 'Billing Staff' (test9@gmail.com) with a 'Sign Out' button. The top right shows the user is logged in as 'Billing Department' (test9@gmail.com).

Figure 11.1: The PMS Billing Staff Dashboard

### 11.1 Centralized Billing Workflow Overview

The system follows a staging-and-consolidation billing architecture:

- Service Delivery:** As patients visit the clinic, receive prescriptions from the Pharmacy, or undergo tests in the Laboratory, each department dispatches line items into the central **Pending Bill Items** staging queue.
- Pending Queue Monitoring:** Billing staff review patients who have unbilled charges accumulated from one or more departments.
- Consolidated Invoicing:** Staff combine pending departmental charges, add optional custom items, apply discounts or tax, and generate a finalized Invoice. Staged items automatically transition from **PENDING** to **BILLED**.
- Payment Collection:** Staff record payments (full or partial) against issued invoices and issue receipts.

## 11.2 Process Pending Bills Panel

Accessible via the “**Process Pending Bills**” sidebar menu:

- **Patients with Pending Bills** — Displays a real-time list of all patients who have unbilled charges awaiting processing.
- **Departmental Item Review** — Selecting a patient displays all staged items grouped by origin:
  - **PHARMACY** — Medications, dosage quantities, unit prices, and line totals.
  - **LAB** — Diagnostic investigations, lab panels, and specimen processing fees.
  - **RECEPTION / CLINICAL** — Doctor consultation charges, specialist fees, or registration costs.
- **Add Additional Line Items** — Staff can add manual charges on the fly (e.g., room charges, medical consumables, emergency care supplements) specifying description, quantity, and unit price.
- **Adjustments & Calculations:**
  - Automatically calculates subtotal from pending items and additional entries.
  - **Discount** — Apply flat discount amounts where applicable (e.g., institutional concessions).
  - **Tax** — Add applicable government/hospital taxes.
  - **Clinical Notes** — Add internal or invoice-facing notes (e.g., insurance claim numbers, payment terms).
- **Generate Invoice** — Clicking “**Generate Invoice**” creates the official invoice with a unique tracking number (e.g., INV-2026-01042), sets the initial status to **ISSUED**, and marks all staged items as **BILLED**.

## 11.3 Invoice Management & Invoices List

Accessible via the “**Billing & Invoices**” sidebar menu:

- **Search & Filter** — Search invoices by invoice number, patient name, or patient ID. Filter invoices by status:
  - **ISSUED** — Invoice generated, awaiting payment.
  - **PARTIALLY\_PAID** — Partial installment received, balance remaining.
  - **PAID** — Fully settled.
  - **CANCELLED** — Voided invoice.
- **Invoice Details & Breakdown** — Inspect line-item charges, tax, discounts, timestamps, creating staff member, and customer payment history.
- **Print / Download** — Generate printable invoice slips and receipts for patients.

## 11.4 Payment Processing

- **Record Payment** — Click “Pay” on any unpaid or partially paid invoice.
- **Payment Amounts** — Accept full payments or partial installments. The system automatically computes the remaining balance:
  - If payment settles the full total, status transitions to PAID.
  - If payment is partial, status updates to PARTIALLY\_PAID.
- **Supported Payment Methods:**
  - Cash
  - Credit / Debit Card
  - Insurance / Third-Party Payer
  - Bank Transfer / Online
- **Audit Logging** — Every invoice created and payment recorded is automatically captured in the immutable audit log for financial compliance.

## 11.5 Revenue Summary & Financial Metrics

Top-level metrics give billing administrators instant visibility into hospital revenue:

- **Total Revenue Collected** — Cumulative monetary value of all processed payments (PAID and PARTIALLY\_PAID).
- **Completed Invoices** — Total count of fully paid customer invoices.
- **Pending Receivables** — Real-time tracking of outstanding hospital balances.

## Chapter 12

# Admin Dashboard

The **Admin Dashboard** provides system administration tools:

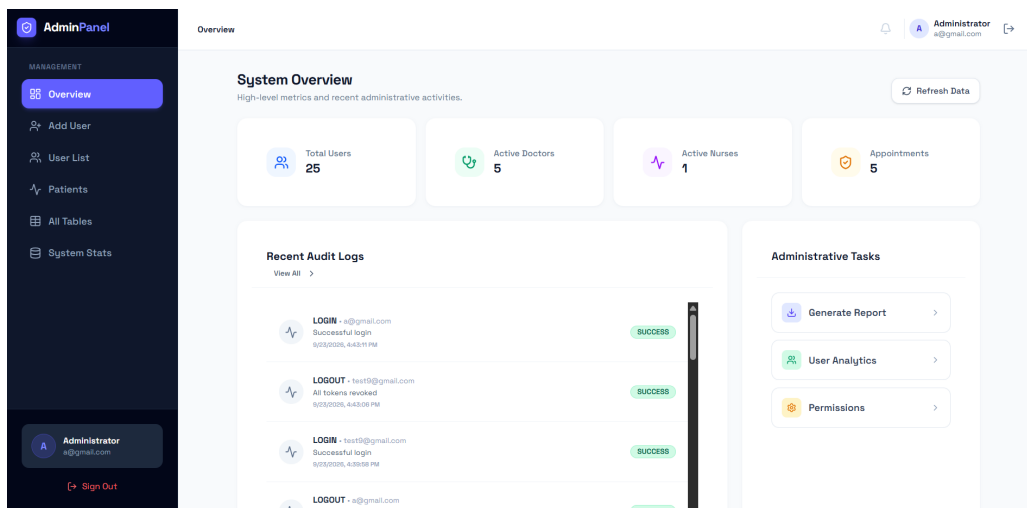


Figure 12.1: The PMS Admin Dashboard

## 12.1 User Management

- View all registered users.
- **Activate/Deactivate** user accounts.
- **Change user roles** as needed.
- Delete user accounts.

## 12.2 Audit Logs

View a comprehensive log of all system activities including:

- User logins and logouts
- Record access and modifications
- Administrative actions

## 12.3 System Reports

Generate administrative and statistical reports.

## Chapter 13

# Management Dashboard

The **Management Dashboard** focuses on organizational oversight:

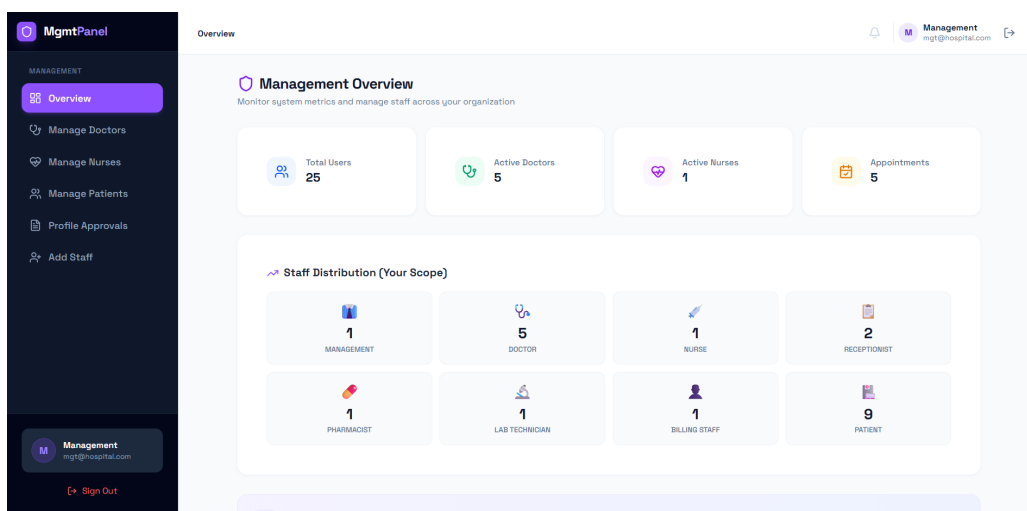


Figure 13.1: The PMS Management Dashboard

### 13.1 Staff Management

View and manage staff accounts (excludes Admin-level accounts). Monitor staff activity and workload.

### 13.2 Operational Reports

Access reports on appointment volumes, patient demographics, and staff utilization.

## Chapter 14

# Notifications

The PMS features a **real-time notification system** powered by WebSocket technology:

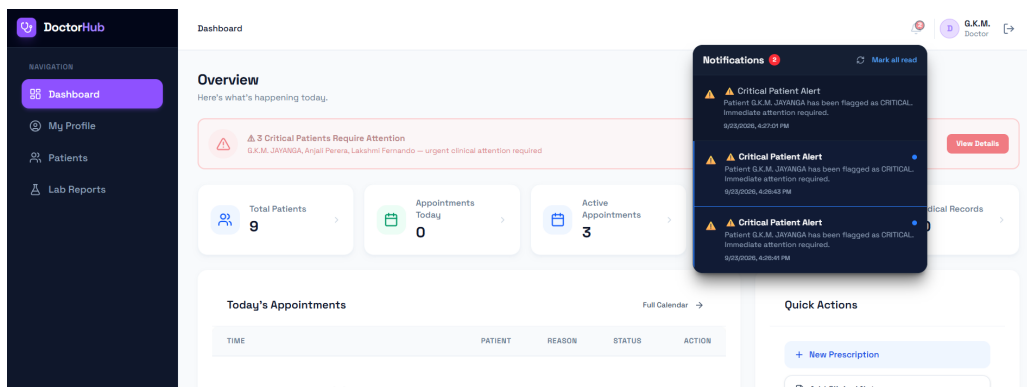


Figure 14.1: The PMS Notification tab

- **Bell Icon** — A notification bell in the navigation bar shows unread notification count.
- Click the bell icon to view recent notifications.
- Notification types include:
  - New appointment bookings
  - Appointment cancellations
  - Lab results availability
  - Prescription updates
  - System announcements

## Chapter 15

# Frequently Asked Questions (FAQ)

### Q: I forgot my password. What do I do?

Contact your system administrator to reset your password.

### Q: Why can't I access certain features?

The system uses role-based access control. You can only access features assigned to your role. Contact your administrator if you need additional permissions.

### Q: My session keeps expiring. Is this normal?

Yes. For security, sessions expire after 15 minutes of inactivity. The system automatically refreshes your session while you are actively using it.

### Q: Can I use the system on my phone?

Yes. The PMS is built with responsive design and works on mobile browsers, tablets, and desktops.

### Q: How is my data protected?

The system uses JWT authentication, encrypted passwords (BCrypt), HTTPS encryption, role-based access, and audit logging to protect your data.

## Chapter 16

# Troubleshooting

| Problem                           | Solution                                                                                                                                        |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cannot log in</b>              | Verify your email and password. Check if your account is active. Your account may be locked after 5 failed attempts (lockout lasts 15 minutes). |
| <b>Page not loading</b>           | Clear your browser cache (Ctrl+Shift+Delete). Try a different browser. Check your internet connection.                                          |
| <b>“Access Denied” error</b>      | You do not have permission for this page. Verify you are logged in with the correct role.                                                       |
| <b>Notifications not updating</b> | Refresh the page. Ensure your browser allows WebSocket connections.                                                                             |
| <b>File upload fails</b>          | Ensure the file is under 10 MB. Supported formats: PDF, JPEG, PNG.                                                                              |
| <b>Seeing outdated data</b>       | Refresh the page (F5). If the issue persists, log out and log back in.                                                                          |

## Chapter 17

# Contact & Support

For technical support or questions, contact the development team:

| Team Member                         | Email                                                          |
|-------------------------------------|----------------------------------------------------------------|
| S.M.L.E Senadhipathi<br>(E/22/364)  | <a href="mailto:e22364@eng.pdn.ac.lk">e22364@eng.pdn.ac.lk</a> |
| D.D.S.K. Gunawardhana<br>(E/22/125) | <a href="mailto:e22125@eng.pdn.ac.lk">e22125@eng.pdn.ac.lk</a> |
| G.K.M. Jayanga (E/22/159)           | <a href="mailto:e22159@eng.pdn.ac.lk">e22159@eng.pdn.ac.lk</a> |
| D.D. Abeysinghe (E/22/004)          | <a href="mailto:e22004@eng.pdn.ac.lk">e22004@eng.pdn.ac.lk</a> |

## Project Links

- **GitHub Repository:** [github.com/cepdnack/e22-2yp-co2060-Patient-Management-System](https://github.com/cepdnack/e22-2yp-co2060-Patient-Management-System)
- **Project Page:** [cepdnack.github.io/e22-co2060-Patient-Management-System](https://cepdnack.github.io/e22-co2060-Patient-Management-System)
- **Department of Computer Engineering:** [ce.pdn.ac.lk](https://ce.pdn.ac.lk)
- **University of Peradeniya:** [eng.pdn.ac.lk](https://eng.pdn.ac.lk)

*Patient Management System — User Manual*  
*Department of Computer Engineering, University of Peradeniya*  
*Last Updated: September 2026*